PK-PD Practice 3

Module 7b - PK-PD Relationship · Daily practice sheet · No date

Name _____ Date _____

A 75-kg patient is to receive a single 20-mg/kg IV bolus dose of an antibiotic. The antibiotic follows a one-compartment open model, has an apparent volume of distribution that is approximately 0.3 L/kg, a first-order elimination half-life of approximately 3.6 hours, and is characterized by a minimum effective concentration of 7.2 mcg/mL.

a What is the expected duration of action of this antibiotic in this patient?

b What is the expected duration of action of this antibiotic in this patient if the dose were doubled?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

a

Her key

Answer: 11.56 hr

WHAT THE STEM GIVES

weight	75-kg	cancels: dose and volume are both per kilogram
D_0	20-mg/kg	over 0.3 L/kg gives C_0 = 66.67 mg/L
V_D	approximately 0.3 L/kg	the denominator of C_0
t½	approximately 3.6 hours	multiplies the logarithm
Ceff	7.2 mcg/mL	equals 7.2 mg/L, the MEC

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_0 = \frac{20\ \text{mg/kg}}{0.3\ \text{L/kg}} = 66.67\ \text{mg/L};\ \text{Ceff} = 7.2\ \text{mcg/mL} = 7.2\ \text{mg/L}$$

Dose and volume are both per kilogram, so the kilograms cancel in C_0 and the 75-kg weight is not needed. 1 mcg/mL equals 1 mg/L.

ALGEBRA

$$\text{teff} = \frac{t½\ \ln(C_0 \div \text{Ceff})}{0.693} = \frac{(3.6\ \text{hr})\ \ln(66.67 \div 7.2)}{0.693} = \frac{(3.6\ \text{hr})\ \ln 9.259}{0.693} = \frac{(3.6\ \text{hr})(2.226)}{0.693}$$

Her key uses the half-life form written as t½ over 0.693, which is 1.44 t½ unrounded. The level must fall 9.26-fold to reach the MEC.

ROUND

$$\text{teff} = 11.56\ \text{hr}$$

8.013 ÷ 0.693 = 11.56 hr, her boxed value: between 3 and 4 half-lives of 3.6 hr, because 9.26 lies between 8 and 16.

> **DOES THE NUMBER MAKE SENSE?**
> C_0 ÷ Ceff = 9.259 is between 8 and 16, so teff is between 3 and 4 half-lives of 3.6 hr: 11.56 hr.

b

Her key

Answer: 15.16 hr

WHAT THE STEM GIVES

D_0	20-mg/kg	doubled to 40 mg/kg, so C_0 doubles
V_D	approximately 0.3 L/kg	unchanged by the dose
t½	approximately 3.6 hours	the increase in teff
Ceff	7.2 mcg/mL	equals 7.2 mg/L

THE WORKING, ONE LINE AT A TIME

SETUP

Doubled dose: 40 mg/kg; $C_0 = \frac{40 \text{ mg/kg}}{0.3 \text{ L/kg}} = 133.3 \text{ mg/L}$; Ceff = 7.2 mg/L

Twice the dose gives twice C_0, because the volume of distribution does not change with the dose. 7.2 mcg/mL equals 7.2 mg/L, the MEC.

ALGEBRA

$$\text{teff} = \frac{(3.6 \text{ hr}) \ln(133.3 \div 7.2)}{0.693} = \frac{(3.6 \text{ hr}) \ln 18.52}{0.693} = \frac{(3.6 \text{ hr})(2.919)}{0.693} = 15.16 \text{ hr}$$

The same half-life form with the doubled C_0. The logarithm grows by ln 2 = 0.693, and 3.6 hr × 0.693 ÷ 0.693 is exactly one half-life, 3.6 hr.

ROUND

15.16 − 11.56 = 3.6 hr, one half-life; teff = 15.16 hr

Her key: "Increase of one half-life", 3.6 hr. Doubling the dose adds one half-life to the duration; it does not double it.

> **DOES THE NUMBER MAKE SENSE?**
> Doubling the dose adds one half-life, 3.6 hr, to 11.56 hr: 15.16 hr, not twice 11.56.